

The Loyola Index Family, Extremal Graph Characterizations and Sensitivity Analysis

Advik Natarajan^a, Ganesh Shiwakoti^b, Michael Arockiaraj^{a*}

^a*Department of Mathematics, Loyola College, Chennai, India*

^b*Department of Mathematics and Statistics, Florida Atlantic University,
USA*

marockiaraj@gmail.com

(Received August 31, 2026)

Abstract

The two-parameter Gutman–Milovanović family $M_{\alpha,\beta}(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta$ unifies the Zagreb, Randić, sum-connectivity and related degree-based indices. We study its normalized degree-imbalance exponential extension, the *Loyola index family*

$$LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta \exp\left(\gamma \frac{|d_u - d_v|}{d_u + d_v}\right),$$

with $LO(G; \alpha, \beta, 0) = M_{\alpha,\beta}(G)$ identically.

We prove that the imbalance direction is affinely independent of the two parent directions on graphs with maximum degree at least 3, that $\log LO$ is globally convex in (α, β, γ) with gradient and Hessian given by the mean and covariance of the edge-feature vector under an induced edge measure, and we extend two-sided bounds from the parent family to $\gamma \neq 0$. On molecular graphs of maximum degree at most four the entire family is confined to the finite-dimensional space of edge-degree-pair counts; we quantify this ceiling exactly, exhibiting octane-isomer pairs that no vertex-degree-based index can distinguish, and showing that the pure- γ points

*Corresponding author.

attain the maximal degree-pair discrimination on trees of orders 7–12. A matched-budget nested leave-one-out comparison against the optimized parent family on five octane properties, repeated over one hundred random-search seeds at two budgets, finds that the free imbalance parameter usually improves out-of-sample prediction on one property and usually degrades it elsewhere, with neither tendency sign-invariant across candidate streams, so γ is presented as a structurally interpretable and provably independent direction rather than a predictive upgrade; under the published structure-sensitivity protocol of Furtula, Gutman and Dehmer, the pure- γ points show the largest sensitivity-to-abruptness ratios in the tested comparison set.

1 Introduction

The literature on degree-based topological indices in chemical graph theory has grown substantially since the seminal Wiener index, the Randić index, and the Zagreb indices of Gutman and Trinajstić. Among the many parametric extensions are the general Randić index R_α [5], the general sum-connectivity index χ_α [38], the forgotten index F revisited by Furtula and Gutman [8], the geometric–arithmetic index GA [35], the Albertson irregularity index [2] and the inverse σ -index problem [14].

While historical literature focused primarily on algebraic formulations to capture molecular branching and structural invariants, a distinct shift toward geometric interpretations occurred when Gutman introduced the Sombor index family [15, 16]. This formulation led to work such as establishing fundamental bounds [25] and evaluating how these new geometric invariants correlated with traditional degree-based metrics [36]. Much of this initial work concentrated on identifying extremal bounds within fundamental graph classes, particularly optimizing the general and standard variants of the index over molecular trees and related networks [1, 6, 7].

Contemporary studies have introduced modified metrics such as the mean Sombor index [24], the elliptic Sombor index [17], and the hyperbolic Sombor index [4]. Concurrently, parameters like the diminished Sombor index [27, 28] and the hybrid vertex-edge KG-Sombor index [22] have been developed to handle unique structural constraints. Beyond purely theoretical analysis, these descriptors are increasingly integrated into practical

quantitative structure-property relationship (QSPR) models and spectral evaluations. For instance, hybrid geometric-harmonic-Zagreb descriptors are now utilized to optimize the predictive analysis of polycyclic aromatic hydrocarbons [3], while modified reverse degree invariants help characterize the graph energy and entropy of complex 2D metal-organic frameworks [20]. This rapid diversification of structural metrics is currently being unified through generalized matrix methods [39], dual coindex formulations [12], simplicial network assessments [33], and finite-dimensional linear frameworks [31], as documented in recent comprehensive literature surveys [18].

The natural two-parameter unification of the product-based and sum-based degree indices is the *Gutman–Milovanović family*, introduced by Gutman, Milovanović and Milovanović [13] and studied intensively in the recent bounds and QSPR literature [11, 26, 29]:

$$M_{\alpha,\beta}(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta, \quad \alpha, \beta \in \mathbb{R}.$$

Its special cases are catalogued in [26]: $M_{0,1} = M_1$ (first Zagreb), $M_{1,0} = M_2$ (second Zagreb), $M_{0,2} = HM$ (hyper-Zagreb), $M_{-1,0} = {}^mM_2$ (modified second Zagreb), $M_{-1/2,0} = R$ (Randić), $M_{0,-1/2} = \chi$ (sum-connectivity [37]), $2M_{0,-1} = H$ (harmonic), $M_{1,-1} = ISI$ (inverse sum indeg), $2M_{1/2,-1} = GA$ (geometric-arithmetic), $\frac{1}{2}M_{-1/2,1} = AG$ (arithmetic-geometric), and the whole general Randić family $R_\alpha = M_{\alpha,0}$ [5] and general sum-connectivity family $\chi_\beta = M_{0,\beta}$ [38] as parameter axes. We claim no novelty for this backbone, for its reduction catalogue, or for its generic tunability: those belong to the Gutman–Milovanović literature.

In this paper we study a three-parameter extension of $M_{\alpha,\beta}$, the *Loyola index family*, obtained by attaching a separately parameterized exponential factor in the normalized degree imbalance $q_{uv} = |d_u - d_v|/(d_u + d_v)$ of each edge: $LO(G; \alpha, \beta, \gamma) = \sum_{uv} (d_u d_v)^\alpha (d_u + d_v)^\beta e^{\gamma q_{uv}}$, so that $LO(G; \alpha, \beta, 0) = M_{\alpha,\beta}(G)$ identically. Positive γ exponentially upweights degree-imbalanced edges, negative γ downweights them, $q_{uv} = 0$ on equal-degree edges, and on regular graphs γ has no effect. The questions this pa-

per answers are therefore about γ alone: is the imbalance direction mathematically independent of the parent parameters, what analytic structure does it carry, and does it add measurable structural or predictive information beyond an optimized Gutman–Milovanović parent?

On the mathematical side, we prove that the answer to independence is affirmative for graphs of maximum degree at least three (Proposition 2), establish global log-convexity of the family in (α, β, γ) with an exact mean/covariance interpretation of its derivatives (Theorem 3), give closed-form values on standard graph classes, and derive two-sided bounds that extend inequalities known for $M_{\alpha, \beta}$ [11, 26] to the $\gamma \neq 0$ regime, several sharp in their equality cases. On the chemometric side we quantify the exact information ceiling of the entire vertex-degree-based class on molecular graphs, run a matched-budget nested comparison of *LO* against its optimized parent, and evaluate the family under the published structure-sensitivity protocol of Furtula, Gutman and Dehmer [9]; no universal ranking or predictive-superiority claim is made.

The distinguishing feature of the Loyola family within the degree-based BID class is the multiplicative $\exp(\gamma q_{uv})$ factor, where $q_{uv} = |d_u - d_v|/(d_u + d_v) \in [0, 1]$ is the normalized degree imbalance. Writing the summand as $\exp(\alpha \ln(d_u d_v) + \beta \ln(d_u + d_v) + \gamma q_{uv})$ shows that the family is a three-parameter subfamily of the exponential vertex-degree-based indices introduced by Rada [30] and studied further in extremal settings by Sigarreta [34]; general sufficient conditions for tree-extremal behaviour of such exponential edge-weight sums are known [10, 19], and no extremal-ordering novelty is claimed here. The published one-parameter exponential indices place their parameter on a single classical generator inside the exponent; the scoped contribution of this family is the multi-parameter *linear combination* in the exponent in which the two Gutman–Milovanović basis terms act as the factors $(d_u d_v)^\alpha (d_u + d_v)^\beta$ while the imbalance coordinate carries its own parameter γ . The same imbalance ratio appears in the diminished Sombor index [28] and in irregularity-based BID variants; to our knowledge, and within the scope of the literature search documented in the reproducibility package, no published index couples it multiplicatively, with its own parameter, to the Gutman–Milovanović family. Since

$0 \leq q_{uv} < 1$, the factor lies in $[1, e^\gamma)$ for $\gamma > 0$, equals 1 for $\gamma = 0$, and lies in $(e^\gamma, 1]$ for $\gamma < 0$; it reduces to 1 on regular graphs. The exponential form is a structural choice rather than one derived from a physical model: it lets γ act multiplicatively and only on irregularity, vanishing on regular graphs.

Also, it is useful to demonstrate the descriptor-space ceiling in which the entire family lives. On hydrogen-suppressed molecular graphs of maximum degree at most 4, every BID index is a linear functional of the ten unordered edge-degree-pair counts $m_{ij}(G) = |\{uv \in E(G) : \{d_u, d_v\} = \{i, j\}\}|$ for $1 \leq i \leq j \leq 4$; this finite-dimensional view of vertex-degree-based indices has recently been placed on a formal linear-algebraic footing by Rada [31]. Thus the Loyola family, like every BID family on this molecular class, lies inside a fixed ten-dimensional descriptor space, no matter how its three real parameters are chosen.

In sum, the contributions of this paper are: (i) the affine independence of the imbalance direction from the two Gutman–Milovanović directions on graphs with $\Delta \geq 3$ (Proposition 2); (ii) global log-convexity of the family with exact mean/covariance formulas for its parameter derivatives (Theorem 3); (iii) two-sided bounds extending parent-family inequalities to $\gamma \neq 0$, several sharp; (iv) an exact quantification of the information ceiling of the whole vertex-degree-based class on molecular graphs, including octane-isomer pairs indistinguishable by every such index (Proposition 4) and floor-attainment of the pure- γ points across tree orders 7–12; and (v) an honest matched-budget nested assessment of whether γ adds out-of-sample predictive information beyond the optimized parent (in a 100-seed analysis it usually helps on one of five octane properties and usually does not elsewhere, with neither tendency sign-invariant across candidate streams), together with the family’s behaviour under the published structure-sensitivity protocol [9], validated against published decane-class values [4]. The family is thus offered not as a universally superior predictor but as a controlled, analytically transparent probe of the BID descriptor space whose third direction is provably independent of its established parent.

2 Preliminaries

2.1 Notation

Let $G = (V, E)$ be a finite simple connected graph of order $n = |V|$ and size $m = |E|$. For a vertex $v \in V$, the degree $d_v = d_G(v)$ counts the edges incident to v . The maximum and minimum degrees are denoted $\Delta = \Delta(G)$ and $\delta = \delta(G)$. In the chemometric sections we work with hydrogen-suppressed molecular graphs, where the assumption $\Delta \leq 4$ is natural for organic molecules built from $\{\text{C}, \text{N}, \text{O}, \text{S}, \text{P}, \text{halogens}\}$.

The classical graph families used throughout are: the complete graph K_n , the cycle C_n , the path P_n , the star S_n , the complete bipartite graph $K_{p,q}$, the hypercube Q_k (the k -dimensional binary hypercube; 2^k vertices, $k \cdot 2^{k-1}$ edges, k -regular), the wheel W_n (an $(n-1)$ -cycle plus a universal vertex), the friendship graph F_p (p triangles sharing a common vertex), and the r -regular graphs.

For an edge $e = uv \in E(G)$ we write (d_u, d_v) for its *degree pair* understood as an unordered multiset $\{d_u, d_v\}$.

2.2 Definition

Definition 1 (Loyola index family). Let $G = (V, E)$ be a simple connected graph. For $\alpha, \beta, \gamma \in \mathbb{R}$, the *Loyola index family* is defined as,

$$LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta \exp\left(\gamma \frac{|d_u - d_v|}{d_u + d_v}\right) \quad (1)$$

Equation (1) is well defined for all $\alpha, \beta \in \mathbb{R}$ on graphs without isolated vertices (so that $d_u, d_v \geq 1$ on every edge). The expression is invariant under permutations of (u, v) because $d_u d_v = d_v d_u$, $d_u + d_v = d_v + d_u$, and $|d_u - d_v| = |d_v - d_u|$; hence $LO(G; \alpha, \beta, \gamma)$ depends only on the multiset of unordered edge degree pairs of G . When $\gamma = 0$ the family reduces identically to the Gutman–Milovanović family [13, 26],

$$LO(G; \alpha, \beta, 0) = M_{\alpha, \beta}(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta,$$

whose classical special cases are catalogued in the Introduction. Throughout, $q_{uv} = |d_u - d_v|/(d_u + d_v)$ denotes the normalized degree imbalance of the edge uv ; positive γ exponentially upweights degree-imbalanced edges, negative γ downweights them, and on regular graphs every $q_{uv} = 0$ so that γ has no effect and $LO(G; \alpha, \beta, \gamma) = M_{\alpha, \beta}(G)$ for all γ .

2.3 Closed-form values on standard graphs

Proposition 1. *Let $\alpha, \beta, \gamma \in \mathbb{R}$. Then:*

1. $LO(K_n; \alpha, \beta, \gamma) = \frac{n(n-1)}{2} 2^\beta (n-1)^{2\alpha+\beta}$ for $n \geq 2$.

2. $LO(C_n; \alpha, \beta, \gamma) = n \cdot 4^{\alpha+\beta}$ for $n \geq 3$.

3. $LO(P_n; \alpha, \beta, \gamma) = \begin{cases} 2^\beta, & n = 2, \\ 2 \cdot 2^\alpha 3^\beta e^{\gamma/3}, & n = 3, \\ 2 \cdot 2^\alpha 3^\beta e^{\gamma/3} + (n-3) \cdot 4^{\alpha+\beta}, & n \geq 4. \end{cases}$

4. $LO(S_n; \alpha, \beta, \gamma) = (n-1)(n-1)^\alpha n^\beta \exp\left(\gamma \frac{n-2}{n}\right)$ for $n \geq 2$.

5. $LO(K_{p,q}; \alpha, \beta, \gamma) = pq(pq)^\alpha (p+q)^\beta \exp\left(\gamma \frac{|p-q|}{p+q}\right)$.

6. $LO(Q_k; \alpha, \beta, \gamma) = k \cdot 2^{k-1} \cdot 2^\beta k^{2\alpha+\beta}$ for $k \geq 1$.

7. For $n \geq 5$, W_n has $n-1$ spoke edges with degree pair $(3, n-1)$ and $n-1$ rim edges with degree pair $(3, 3)$, so

$$LO(W_n; \alpha, \beta, \gamma) = (n-1) \left[3^\alpha (n-1)^\alpha (n+2)^\beta \exp\left(\gamma \frac{n-4}{n+2}\right) + 9^\alpha 6^\beta \right].$$

8. For $p \geq 1$, F_p has $2p$ spoke edges $(2, 2p)$ and p outer edges $(2, 2)$, so

$$LO(F_p; \alpha, \beta, \gamma) = p \cdot 4^{\alpha+\beta} + 2p(4p)^\alpha (2p+2)^\beta \exp\left(\gamma \frac{p-1}{p+1}\right).$$

9. For an r -regular graph G with m edges, $LO(G; \alpha, \beta, \gamma) = m \cdot 2^\beta r^{2\alpha+\beta}$.

Proof. Each case is obtained by enumerating the edge degree pairs of the graph and applying Definition 1. For r -regular graphs and for K_n, C_n, Q_k (all regular), every edge has $d_u = d_v = r$, so $\text{imb} = 0$ and $(d_u d_v)^\alpha (d_u +$

$d_v)^\beta = r^{2\alpha}(2r)^\beta = 2^\beta r^{2\alpha+\beta}$. Items (1), (2), (6), and (9) follow immediately. For S_n all $n-1$ edges have degree pair $(1, n-1)$ with $\text{imb} = (n-2)/n$; item (4) follows. For $K_{p,q}$ all pq edges have degree pair (p, q) ; item (5) follows. For P_n ($n \geq 4$) there are exactly two pendant edges with pair $(1, 2)$ and $n-3$ interior edges with pair $(2, 2)$; item (3) follows by direct summation. The boundary cases $n = 2, 3$ are immediate. Items (7) and (8) are obtained by partitioning the edges of W_n and F_p into their two edge-degree-pair classes respectively and summing. ■

3 Parameter geometry and the role of the imbalance parameter

This section establishes the two structural facts that justify introducing the third parameter at all: the imbalance direction cannot be reproduced by the parent parameters, and the family has a globally convex logarithm whose derivatives have an exact probabilistic meaning. Throughout, write $z_{uv} = (\ln(d_u d_v), \ln(d_u + d_v), q_{uv}) \in \mathbb{R}^3$ for the feature vector of the edge uv and $\theta = (\alpha, \beta, \gamma)$, so that $LO_G(\theta) = \sum_{uv \in E(G)} e^{\theta^\top z_{uv}}$.

Proposition 2 (Affine independence of the imbalance direction). *Let $\Delta \geq 3$ and let $P_\Delta = \{(i, j) : 1 \leq i \leq j \leq \Delta\}$ be the admissible degree pairs. Then the augmented feature vectors $(1, \ln(ij), \ln(i+j), q_{ij})$, $(i, j) \in P_\Delta$, span $\mathbb{R}^4$. Consequently, if two parameter triples $\theta \neq \theta'$ satisfy $e^{\theta^\top z} = c e^{\theta'^\top z}$ for some constant $c > 0$ and all $(i, j) \in P_\Delta$, then $\theta = \theta'$ and $c = 1$; in particular no choice of (α', β') and scaling constant reproduces the action of $\gamma \neq 0$ on all admissible degree pairs. For $\Delta = 2$ the span has dimension 3, and for $\Delta = 1$ (the single pair $(1, 1)$, augmented vector $(1, 0, \ln 2, 0)$) it has dimension 1; in both cases the conclusion fails.*

Proof. It suffices to show that q_{ij} is not an affine function of $(\ln(ij), \ln(i+j))$ on P_Δ ; the four pairs $(1, 1), (2, 2), (1, 2), (1, 3)$, all admissible once $\Delta \geq 3$, witness this. Suppose $q_{ij} = a + b \ln(ij) + c \ln(i+j)$ on these pairs. From $(1, 1)$ and $(2, 2)$, where $q = 0$: $a + c \ln 2 = 0$ and $a + (2b + 2c) \ln 2 = 0$, giving $a = -c \ln 2$ and $b = -c/2$. Substituting into the $(1, 2)$ equation $\frac{1}{3} = a + b \ln 2 + c \ln 3$ yields $c = \frac{1}{3} (\ln 3 - \frac{3}{2} \ln 2)^{-1}$, and then the $(1, 3)$ equation would

require $\frac{1}{2} = c(\ln 2 - \frac{1}{2} \ln 3) \approx 0.814$, a contradiction. Hence the augmented vectors have rank 4. If $e^{\theta^\top z} = c e^{\theta'^\top z}$ on P_Δ , then $(\theta - \theta')^\top z_{ij} = \ln c$ for all admissible pairs, i.e. $(\ln c, \theta' - \theta)$ is orthogonal to every augmented vector, so $\theta = \theta'$ and $c = 1$. For $\Delta = 2$ only the pairs $(1, 1), (1, 2), (2, 2)$ exist and a direct computation shows the augmented vectors have rank 3; for $\Delta = 1$ the single augmented vector has rank 1. $\blacksquare$

Remark (Independence versus per-graph identifiability). Proposition 2 is a statement about the edge-weight *function* on the full admissible pair set, not about every individual graph: on a single graph G , the parameters are identifiable (up to the scaling noted above) only when the degree pairs actually realised by G carry four affinely independent augmented feature vectors. On a regular graph nothing about θ is identifiable beyond one linear functional; on a graph whose edges realise at most three affinely independent augmented vectors, some direction of θ — possibly the γ direction — is unidentifiable from $LO(G; \cdot)$ alone. The proposition guarantees that no such degeneracy is forced on the whole molecular class $\Delta \geq 3$: the family separates parameters as a family of descriptors, even though particular graphs may not.

Theorem 3 (Global log-convexity). *Fix a graph G with $m \geq 1$ edges and define $F_G(\theta) = \log LO_G(\theta)$ on $\mathbb{R}^3$. Let $\mathbb{E}_\theta$ and Cov_θ denote mean and covariance under the edge probability measure $w_{uv}(\theta) = e^{\theta^\top z_{uv}} / LO_G(\theta)$. Then F_G is convex, with*

$$\nabla F_G(\theta) = \mathbb{E}_\theta[z], \quad \nabla^2 F_G(\theta) = \text{Cov}_\theta(z) \succeq 0.$$

In particular $\partial_\gamma F_G = \mathbb{E}_\theta[q] \in [0, 1)$ and $\partial_\gamma^2 F_G = \text{Var}_\theta(q) \geq 0$, so $\log LO$ is nondecreasing and convex in γ , strictly increasing in γ unless every edge of G joins vertices of equal degree. Moreover $\text{Var}_\theta(q) = 0$ at some (equivalently, every) θ if and only if q_{uv} is constant over the edges of G : this covers regular graphs ($q \equiv 0$), but also, e.g., (Δ, δ) -biregular graphs, on which every edge carries the same nonzero imbalance and $\log LO$ is affine and strictly increasing in γ . F_G is strictly convex if and only if the edge feature vectors $\{z_{uv}\}_{uv \in E}$ do not lie in a proper affine subspace of $\mathbb{R}^3$; on a regular graph $F_G(\theta) = \log m + \theta^\top z_0$ is affine and $\text{Cov}_\theta(z) = 0$.

Proof. Direct differentiation of $LO_G(\theta) = \sum_{uv} e^{\theta^\top z_{uv}}$ gives $\partial_k F_G = \sum_{uv} w_{uv} z_{uv,k} = \mathbb{E}_\theta[z_k]$ and $\partial_k \partial_l F_G = \mathbb{E}_\theta[z_k z_l] - \mathbb{E}_\theta[z_k] \mathbb{E}_\theta[z_l]$, the (k, l) -entry of $\text{Cov}_\theta(z)$, which is positive semidefinite; convexity follows. The measure $w(\theta)$ has full support on E for every θ , so $\text{Cov}_\theta(z)$ is singular in a direction v iff $v^\top z_{uv}$ is constant over all edges, i.e. iff the z_{uv} lie in a proper affine subspace; this proves the strictness criterion. Since $q_{uv} \in [0, 1)$, $\mathbb{E}_\theta[q] \in [0, 1)$, and $\mathbb{E}_\theta[q] = 0$ iff $q_{uv} = 0$ on every edge. On an r -regular graph all z_{uv} equal $z_0 = (2 \ln r, \ln 2r, 0)$. ■

Theorem 3 places the family in the classical exponential-family framework: moving γ reweights the edge measure toward imbalanced edges at rate $\mathbb{E}_\theta[q]$, and the Cauchy–Schwarz midpoint inequality of Section 4 is precisely midpoint log-convexity, i.e. a corollary of this theorem.

Proposition 4 (Degree-pair information ceiling). *If two graphs G and H have identical unordered edge-degree-pair count vectors, $m_{ij}(G) = m_{ij}(H)$ for all $i \leq j$, then $LO(G; \theta) = LO(H; \theta)$ for every $\theta \in \mathbb{R}^3$, and more generally $TI(G) = TI(H)$ for every vertex-degree-based index TI . Among the 18 octane isomers exactly 16 distinct count vectors occur: the pairs $\{3\text{-methylheptane}, 4\text{-methylheptane}\}$ and $\{3,4\text{-dimethylhexane}, 2\text{-methyl-3-ethylpentane (alphabetized IUPAC form: 3-ethyl-2-methylpentane)}\}$ share their count vectors while differing in measured properties (Table 2), so every vertex-degree-based descriptor, tuned or not, carries a nonzero irreducible error on this benchmark.*

Proof. By Definition 1, $LO(G; \theta)$ depends on G only through the multiset of unordered edge degree pairs, i.e. through $(m_{ij}(G))_{i \leq j}$; the same holds for every vertex-degree-based index [31]. The counts for the 18 octane isomers are computed and cross-checked in the reproducibility package; both collision pairs have their common count vectors listed in Table 2. ■

The ceiling is tight in the following sense: distinct count vectors are separated by the family at almost every parameter choice.

Proposition 5 (Generic discrimination of distinct profiles). *Fix $\Delta \geq 1$ and let $m \neq m'$ be two distinct edge-degree-pair count vectors supported on the admissible pairs P_Δ . Then $D(\theta) = \sum_{(i,j) \in P_\Delta} (m_{ij} - m'_{ij}) e^{\theta^\top z_{ij}}$ is a*

real-analytic function of θ that is not identically zero, and its zero set has Lebesgue measure zero in $\mathbb{R}^3$. Consequently, any two graphs with distinct count vectors are distinguished by $LO(\cdot; \theta)$ for almost every $\theta \in \mathbb{R}^3$; only graphs with identical count vectors (Proposition 4) are indistinguishable for all θ .

Proof. The map $(i, j) \mapsto z_{ij}$ is injective on P_Δ , since $\ln(ij)$ and $\ln(i + j)$ recover the product and the sum, which determine the unordered pair. Choose $v \in \mathbb{R}^3$ such that the inner products $v^\top z_{ij}$ are pairwise distinct (this excludes finitely many hyperplanes of v 's). Along the ray $\theta = tv$, $D(tv) = \sum_k c_k e^{t\lambda_k}$ with distinct exponents λ_k and coefficients $c_k = m_{i_k j_k} - m'_{i_k j_k}$, not all zero; such an exponential sum vanishes identically only if all $c_k = 0$, so $D \not\equiv 0$ on the ray and hence on $\mathbb{R}^3$. D is a finite sum of exponentials of linear functionals, hence real-analytic on $\mathbb{R}^3$, and the zero set of a nonzero real-analytic function has Lebesgue measure zero. ■

4 Bounds for the Loyola family

This section establishes a coherent set of upper and lower bounds on $LO(G; \alpha, \beta, \gamma)$. The development proceeds from a bracket bound via the edge-contribution function ψ , bounds in (n, m, Δ, δ) for the non-negative parameter region, Cauchy–Schwarz and Jensen-type bounds in the first two Zagreb indices M_1 and M_2 , a uniform ratio-bound theorem, and bounds via the generalised Randić, sum-connectivity, and Sombor indices. Throughout, G is a simple connected graph with $\delta \leq d_v \leq \Delta$ for every $v \in V(G)$.

Remark (Attribution at $\gamma = 0$). At $\gamma = 0$ the family coincides with $M_{\alpha, \beta}$, for which inequalities of all the types developed here are already known: (Δ, δ, m) -bounds with regular/biregular equality and two-sided comparisons against M_1 , M_2 , R , χ , H , GA , AG and ISI appear in [26], and Jensen-, Hölder- and Cauchy–Schwarz-type relations in [11, 29]. The $\gamma = 0$ specializations of the results below are therefore not claimed as new; the content of this section is their extension to $\gamma \neq 0$, where the imbalance factor $e^{\gamma q_{uv}}$ enters every edge weight, and the tracking of how γ deforms

the equality cases.

The summand in (1) depends only on the unordered edge degree pair (d_u, d_v) . We isolate this dependence as follows:

Definition 2. For real parameters α, β, γ and positive reals x, y ,

$$\psi(x, y; \alpha, \beta, \gamma) = (xy)^\alpha (x + y)^\beta \exp\left(\gamma \frac{|x - y|}{x + y}\right). \quad (2)$$

With this notation $LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} \psi(d_u, d_v; \alpha, \beta, \gamma)$.

Proposition 6. *Let G be a simple connected graph of size $m \geq 1$ with degrees in $[\delta, \Delta]$. Then*

$$m \cdot \psi_{\min}(\delta, \Delta; \alpha, \beta, \gamma) \leq LO(G; \alpha, \beta, \gamma) \leq m \cdot \psi_{\max}(\delta, \Delta; \alpha, \beta, \gamma), \quad (3)$$

where, $\psi_{\min} = \min\{\psi(i, j; \alpha, \beta, \gamma) : \delta \leq i \leq j \leq \Delta\}$ and $\psi_{\max} = \max\{\psi(i, j; \alpha, \beta, \gamma) : \delta \leq i \leq j \leq \Delta\}$. The upper inequality is an equality if and only if every edge of G has a degree pair attaining $\psi_{\max}$, and the lower inequality is an equality if and only if every edge has a degree pair attaining $\psi_{\min}$.

Proof. Because G has $\delta \leq d_v \leq \Delta$ for every $v \in V(G)$, every edge $uv \in E(G)$ has its degree pair $\{d_u, d_v\}$ in the admissible set $\mathcal{P}(\delta, \Delta) := \{(i, j) \in \mathbb{Z}^2 : \delta \leq i \leq j \leq \Delta\}$. By the definitions of $\psi_{\min}$ and $\psi_{\max}$ as the minimum and maximum of ψ over $\mathcal{P}(\delta, \Delta)$,

$$\psi_{\min}(\delta, \Delta; \alpha, \beta, \gamma) \leq \psi(d_u, d_v; \alpha, \beta, \gamma) \leq \psi_{\max}(\delta, \Delta; \alpha, \beta, \gamma).$$

Summing each of the two per-edge inequalities over the m edges of G and using $\sum_{uv \in E(G)} \psi(d_u, d_v; \alpha, \beta, \gamma) = LO(G; \alpha, \beta, \gamma)$ gives

$$m \cdot \psi_{\min}(\delta, \Delta; \alpha, \beta, \gamma) \leq LO(G; \alpha, \beta, \gamma) \leq m \cdot \psi_{\max}(\delta, \Delta; \alpha, \beta, \gamma),$$

For the equality cases, the upper inequality $LO(G; \alpha, \beta, \gamma) \leq m \cdot \psi_{\max}$ is the sum of m per-edge inequalities $\psi(d_u, d_v; \alpha, \beta, \gamma) \leq \psi_{\max}$; this sum is an equality if and only if every per-edge inequality is, equivalently every

edge uv has a degree pair attaining $\psi_{\max}$ over $\mathcal{P}(\delta, \Delta)$. The lower equality case is the dual. $\blacksquare$

Theorem 7. *Let $\alpha_1 + \alpha_2 = 2\alpha$, $\beta_1 + \beta_2 = 2\beta$, and $\gamma_1 + \gamma_2 = 2\gamma$. Then for every simple connected graph G without isolated vertices,*

$$LO(G; \alpha, \beta, \gamma) \leq \sqrt{LO(G; \alpha_1, \beta_1, \gamma_1) \cdot LO(G; \alpha_2, \beta_2, \gamma_2)}$$

Equality holds if and only if the ratio $\psi(d_u, d_v; \alpha_1, \beta_1, \gamma_1)/\psi(d_u, d_v; \alpha_2, \beta_2, \gamma_2)$ is the same for every edge of G .

Proof. For each edge $e = uv \in E(G)$ and each $k \in \{1, 2\}$ write

$$\psi_e^{(k)} := \psi(d_u, d_v; \alpha_k, \beta_k, \gamma_k) = (d_u d_v)^{\alpha_k} (d_u + d_v)^{\beta_k} \exp\left(\gamma_k \frac{|d_u - d_v|}{d_u + d_v}\right).$$

Both $\psi_e^{(1)}$ and $\psi_e^{(2)}$ are strictly positive on every edge of a graph without isolated vertices.

By the additivity of the exponents and the hypothesis $\alpha_1 + \alpha_2 = 2\alpha$, $\beta_1 + \beta_2 = 2\beta$, $\gamma_1 + \gamma_2 = 2\gamma$,

$$\begin{aligned} \psi_e^{(1)} \psi_e^{(2)} &= (d_u d_v)^{\alpha_1 + \alpha_2} (d_u + d_v)^{\beta_1 + \beta_2} \exp\left((\gamma_1 + \gamma_2) \frac{|d_u - d_v|}{d_u + d_v}\right) \\ &= (d_u d_v)^{2\alpha} (d_u + d_v)^{2\beta} \exp\left(2\gamma \frac{|d_u - d_v|}{d_u + d_v}\right) \\ &= \psi(d_u, d_v; \alpha, \beta, \gamma)^2. \end{aligned}$$

Taking square roots gives $\sqrt{\psi_e^{(1)} \psi_e^{(2)}} = \psi(d_u, d_v; \alpha, \beta, \gamma)$ for every edge. Apply the Cauchy–Schwarz inequality to the positive sequences $a_e = \sqrt{\psi_e^{(1)}}$, $b_e = \sqrt{\psi_e^{(2)}}$ indexed over $E(G)$:

$$\sum_{e \in E(G)} a_e b_e \leq \sqrt{\sum_{e \in E(G)} a_e^2} \sqrt{\sum_{e \in E(G)} b_e^2}.$$

Substituting $a_e b_e = \sqrt{\psi_e^{(1)} \psi_e^{(2)}} = \psi(d_u, d_v; \alpha, \beta, \gamma)$, $\sum_e a_e^2 = \sum_e \psi_e^{(1)} =$

$LO(G; \alpha_1, \beta_1, \gamma_1)$, and $\sum_e b_e^2 = \sum_e \psi_e^{(2)} = LO(G; \alpha_2, \beta_2, \gamma_2)$ gives

$$\begin{aligned} LO(G; \alpha, \beta, \gamma) &= \sum_{e \in E(G)} \psi(d_u, d_v; \alpha, \beta, \gamma) \\ &\leq \sqrt{LO(G; \alpha_1, \beta_1, \gamma_1) LO(G; \alpha_2, \beta_2, \gamma_2)}, \end{aligned}$$

which is the inequality.

The Cauchy-Schwarz inequality is an equality if and only if the sequences (a_e) and (b_e) are proportional, i.e. there exists a constant $\lambda \geq 0$ such that $b_e = \lambda a_e$ for every $e \in E(G)$, equivalently $\psi_e^{(2)}/\psi_e^{(1)} = \lambda^2$ is the same constant on every edge. ■

Proposition 8. *Let G be a graph of size m and maximum degree Δ . For $\alpha, \beta \geq 0$ and $\gamma \geq 0$,*

$$LO(G; \alpha, \beta, \gamma) \leq m \cdot \Delta^{2\alpha} \cdot (2\Delta)^\beta \cdot \exp\left(\gamma \frac{\Delta - 1}{\Delta + 1}\right). \quad (4)$$

Proof. Fix an edge $uv \in E(G)$. By the definition of Δ we have $1 \leq d_u, d_v \leq \Delta$. We bound the three multiplicative factors of $\psi(d_u, d_v; \alpha, \beta, \gamma)$ separately.

First factor. Because $d_u, d_v \leq \Delta$ we obtain $d_u d_v \leq \Delta^2$. Since $\alpha \geq 0$ and the map $t \mapsto t^\alpha$ is non-decreasing on $\mathbb{R}_{>0}$, $(d_u d_v)^\alpha \leq \Delta^{2\alpha}$.

Second factor. Similarly $d_u + d_v \leq 2\Delta$, and the map $t \mapsto t^\beta$ is non-decreasing for $\beta \geq 0$, so $(d_u + d_v)^\beta \leq (2\Delta)^\beta$.

Third factor. The normalised imbalance $\text{imb}(x, y) = |x - y|/(x + y)$ is maximised over the admissible set $\{(i, j) : 1 \leq i \leq j \leq \Delta\}$ at the pair $(1, \Delta)$, where $\text{imb}(1, \Delta) = (\Delta - 1)/(\Delta + 1)$. Hence $|d_u - d_v|/(d_u + d_v) \leq (\Delta - 1)/(\Delta + 1)$ for every edge. Because $\gamma \geq 0$ and $\exp$ is non-decreasing, $\exp(\gamma |d_u - d_v|/(d_u + d_v)) \leq \exp(\gamma(\Delta - 1)/(\Delta + 1))$. Multiplying the three factor-wise bounds and summing over the m edges of G yields (4). We note that the three per-factor maxima are not simultaneously attainable for $\gamma > 0$, so the bound, while valid, is not sharp in that range. ■

Proposition 9. *Let G be a graph of size m and minimum degree δ . For*

$\alpha, \beta \geq 0$ and $\gamma \geq 0$,

$$LO(G; \alpha, \beta, \gamma) \geq m \cdot \delta^{2\alpha} \cdot (2\delta)^\beta, \quad (5)$$

with equality if and only if G is δ -regular, except in the degenerate case $(\alpha, \beta, \gamma) = (0, 0, 0)$, where $LO(G; \alpha, \beta, \gamma) \equiv m$ and the bound holds with equality for every graph.

Proof. Fix an arbitrary edge $uv \in E(G)$. By the definition of $\delta = \delta(G)$ we have $d_u, d_v \geq \delta$, and we bound the three multiplicative factors of $\psi(d_u, d_v; \alpha, \beta, \gamma)$ separately. Because $d_u, d_v \geq \delta$ and $\alpha \geq 0$, $(d_u d_v)^\alpha \geq \delta^{2\alpha}$; similarly $(d_u + d_v)^\beta \geq (2\delta)^\beta$; and since $\gamma \geq 0$ and the imbalance is non-negative, $\exp(\gamma |d_u - d_v| / (d_u + d_v)) \geq 1$. Multiplying the three lower bounds gives $\psi(d_u, d_v; \alpha, \beta, \gamma) \geq \delta^{2\alpha} (2\delta)^\beta$ for every edge, and summing over the m edges yields (5).

For the equality case, if $(\alpha, \beta, \gamma) = (0, 0, 0)$ then $LO(G; 0, 0, 0) = m$ and the lower bound is also m , so equality holds for every graph. Assume now that $(\alpha, \beta, \gamma) \neq (0, 0, 0)$. Equality in the summed inequality requires equality in every active per-edge factor. If $\alpha > 0$ or $\beta > 0$, this forces $d_u = d_v = \delta$ on every edge. If $\alpha = \beta = 0$ and $\gamma > 0$, equality in the exponential factor requires $d_u = d_v$ on every edge; since G is connected, this implies that G is regular, hence δ -regular. Conversely, if G is δ -regular then every edge has degree pair (δ, δ) and equality holds. Thus, outside the degenerate case $(0, 0, 0)$, equality holds if and only if G is δ -regular. ■

Theorem 10. *Let $\alpha = \gamma = 0$ and $\beta \geq 1$. Then*

$$LO(G; 0, \beta, 0) = \sum_e (d_u + d_v)^\beta \geq m^{1-\beta} M_1(G)^\beta,$$

where $M_1(G) = \sum_{v \in V} d_v^2 = \sum_{uv \in E} (d_u + d_v)$. At $\beta = 1$ both sides equal $M_1(G)$ identically; for $\beta > 1$ equality holds if and only if $d_u + d_v$ is constant on every edge of G .

Proof. Setting $\alpha = \gamma = 0$ in Definition 1 collapses the edge-contribution function to $\psi(d_u, d_v; 0, \beta, 0) = (d_u + d_v)^\beta$, so $LO(G; 0, \beta, 0) = \sum_{uv \in E(G)} (d_u + d_v)^\beta$. The function $\varphi(t) = t^\beta$ satisfies $\varphi''(t) =$

$\beta(\beta - 1)t^{\beta-2} \geq 0$ for $\beta \geq 1$; hence φ is convex on $\mathbb{R}_{>0}$, with strict convexity for $\beta > 1$. Apply Jensen's inequality (convex case) to the $m = |E(G)|$ positive values $\{d_u + d_v : uv \in E(G)\}$ with uniform weights $1/m$:

$$\left(\frac{M_1(G)}{m}\right)^\beta = \varphi\left(\frac{1}{m} \sum_{uv} (d_u + d_v)\right) \leq \frac{1}{m} \sum_{uv} (d_u + d_v)^\beta = \frac{LO(G; 0, \beta, 0)}{m},$$

using $\sum_{uv \in E(G)} (d_u + d_v) = M_1(G)$. Multiplying both sides by $m > 0$ yields the bound. At $\beta = 1$, φ is affine and Jensen is the identity $LO(G; 0, 1, 0) = M_1(G)$; for $\beta > 1$ strict convexity gives equality if and only if $d_u + d_v$ is constant over every edge. ■

Theorem 11. *Let $\alpha = \gamma = 0$ and $0 < \beta < 1$. Then $LO(G; 0, \beta, 0) = \sum_e (d_u + d_v)^\beta \leq m^{1-\beta} M_1(G)^\beta$, with equality if and only if $d_u + d_v$ is constant on every edge of G .*

Proof. The function $t \mapsto t^\beta$ is strictly concave on $\mathbb{R}_{>0}$ for $\beta \in (0, 1)$. Jensen's inequality in the concave case applied to the values $\{d_u + d_v : uv \in E(G)\}$ gives $\frac{1}{m} \sum_e (d_u + d_v)^\beta \leq (M_1(G)/m)^\beta$, using $M_1(G) = \sum_{uv} (d_u + d_v)$. Multiplying by m gives the upper bound; strict concavity gives equality if and only if $d_u + d_v$ is constant on every edge. ■

Theorem 12. *Let $\beta = \gamma = 0$ and $\alpha \geq 1$. Then*

$$LO(G; \alpha, 0, 0) = \sum_{uv \in E(G)} (d_u d_v)^\alpha = R_\alpha(G) \geq m^{1-\alpha} M_2(G)^\alpha,$$

where $M_2(G) = \sum_{uv \in E(G)} d_u d_v$ is the second Zagreb index and $R_\alpha(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha$ is the general Randić index. At $\alpha = 1$ both sides equal $M_2(G)$ identically; for $\alpha > 1$ equality holds if and only if the product $d_u d_v$ is constant on every edge of G .

Proof. Setting $\beta = \gamma = 0$ collapses the edge contribution to $(d_u d_v)^\alpha$, so $LO(G; \alpha, 0, 0) = R_\alpha(G)$. The function $\varphi(t) = t^\alpha$ has $\varphi''(t) = \alpha(\alpha - 1)t^{\alpha-2} \geq 0$ for $\alpha \geq 1$ and is convex, strictly for $\alpha > 1$. Jensen (convex case) on $\{d_u d_v : uv \in E(G)\}$ with weights $1/m$ gives $(M_2(G)/m)^\alpha \leq R_\alpha(G)/m$,

using $\sum_{uv} d_u d_v = M_2(G)$. Multiplying by m yields the bound; $\alpha = 1$ is the affine identity, and for $\alpha > 1$ strict convexity gives equality if and only if $d_u d_v$ is constant on every edge. ■

Proposition 13. *Let $\gamma = 0$, and define $R_a(G) = \sum_{uv \in E(G)} (d_u d_v)^a$ and $\chi_b(G) = \sum_{uv \in E(G)} (d_u + d_v)^b$ (so $R = R_{-1/2}$ and $\chi = \chi_{-1/2}$). Then for every graph G ,*

$$LO(G; \alpha, \beta, 0) \leq \sqrt{R_{2\alpha}(G) \cdot \chi_{2\beta}(G)}. \quad (6)$$

In particular, at $(\alpha, \beta) = (-1/4, -1/4)$ this specialises to

$$LO(G; -1/4, -1/4, 0) \leq \sqrt{R(G) \chi(G)}.$$

Proof. Apply Theorem 7 with $(\alpha_1, \beta_1, \gamma_1) = (2\alpha, 0, 0)$ and $(\alpha_2, \beta_2, \gamma_2) = (0, 2\beta, 0)$, whose components sum to $(2\alpha, 2\beta, 0)$, meeting the hypotheses since $\gamma = 0$. By Definition 1, $LO(G; 2\alpha, 0, 0) = R_{2\alpha}(G)$ and $LO(G; 0, 2\beta, 0) = \chi_{2\beta}(G)$, so Theorem 7 gives (6). The equality case is that of Theorem 7: the ratio $(d_u d_v)^{2\alpha} / (d_u + d_v)^{2\beta}$ is constant over every edge. ■

Proposition 14. *For $\alpha = 0$, $0 < \beta < 1$, and $\gamma = 0$, the Loyola quantity $LO(G; 0, \beta, 0) = \chi_\beta(G)$ satisfies*

$$LO(G; 0, \beta, 0) \leq m^{1-\beta} \cdot (\sqrt{2} SO(G))^\beta, \quad (7)$$

where $SO(G) = \sum_{uv \in E(G)} \sqrt{d_u^2 + d_v^2}$ is the Sombor index.

Proof. For positive reals x, y , $x^2 + y^2 \geq (x + y)^2 / 2$ follows from $(x + y)^2 \leq 2(x^2 + y^2)$ by the AM–GM inequality $xy \leq (x^2 + y^2) / 2$. Taking square roots with $x = d_u, y = d_v$ and summing over $E(G)$ gives $SO(G) \geq M_1(G) / \sqrt{2}$, i.e. $M_1(G) \leq \sqrt{2} SO(G)$. For $0 < \beta < 1$, the concave Jensen bound (Theorem 11) gives $LO(G; 0, \beta, 0) \leq m^{1-\beta} M_1(G)^\beta$, and since $t \mapsto t^\beta$ is non-decreasing for $\beta > 0$, substituting $M_1(G) \leq \sqrt{2} SO(G)$ yields (7). ■

The next theorem gives a single uniform template for bounding $LO(G; \alpha, \beta, \gamma)$ both above and below in terms of any classical bond-

incident-degree index whose edge contribution is strictly positive on the admissible degree pairs of the graph.

Theorem 15. *Let G be a simple graph with maximum degree at most Δ , and let $I_h(G) = \sum_{uv \in E(G)} h(d_u, d_v)$ be a BID index with edge contribution $h(i, j)$ strictly positive on every degree pair (i, j) that appears in G . Define*

$$c_{\min}^h = \min_{(i,j)} \frac{\psi(i, j; \alpha, \beta, \gamma)}{h(i, j)}, \quad c_{\max}^h = \max_{(i,j)} \frac{\psi(i, j; \alpha, \beta, \gamma)}{h(i, j)},$$

where the extrema range over the degree pairs actually appearing in G . Then

$$c_{\min}^h \cdot I_h(G) \leq LO(G; \alpha, \beta, \gamma) \leq c_{\max}^h \cdot I_h(G), \quad (8)$$

with equality on either side if and only if every edge of G attains the corresponding extremum of ψ/h .

Theorem 15 is a *sharp graph-specific* comparison: its constants depend on the degree pairs that actually appear in G . A uniform version follows immediately by enlarging the range of the extrema.

Corollary (Uniform Δ -bounded version). *With the hypotheses of Theorem 15, but with h strictly positive on all of $P_\Delta = \{(i, j) : 1 \leq i \leq j \leq \Delta\}$ and the extrema taken over P_Δ , inequality (8) holds simultaneously for every graph of maximum degree at most Δ , with constants independent of the individual graph. Equality on either side holds if and only if every edge of G realises a degree pair attaining the corresponding extremum over P_Δ ; the uniform constants are in general strictly worse than the graph-specific ones, since the extrema are taken over a superset.*

Proof. Fix an edge $uv \in E(G)$ and define $r_{uv} := \psi(d_u, d_v; \alpha, \beta, \gamma)/h(d_u, d_v) > 0$. By definition $c_{\min}^h \leq r_{uv} \leq c_{\max}^h$, so multiplying by $h(d_u, d_v) > 0$ gives the per-edge bounds $c_{\min}^h h(d_u, d_v) \leq \psi(d_u, d_v; \alpha, \beta, \gamma)$ and $\psi(d_u, d_v; \alpha, \beta, \gamma) \leq c_{\max}^h h(d_u, d_v)$. Summing over $E(G)$ and using $\sum_{uv} h(d_u, d_v) = I_h(G)$ and $\sum_{uv} \psi = LO(G; \alpha, \beta, \gamma)$ gives (8); the equality conditions are that every edge attains the corresponding extremum of ψ/h over the appearing pairs. ■

Proposition 16. *At $\gamma = 0$, $LO(G; \alpha, \beta, 0)$ coincides with the two-parameter (α, β) degree-based index $\sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta$. Let $E_{\max}(G)$ denote the set of edges realising $\text{imb}_{\max}(G) = \max_{uv} \text{imb}(d_u, d_v)$ and $E_{\min}(G)$ the set realising $\text{imb}_{\min}(G) = \min_{uv} \text{imb}(d_u, d_v)$. Then as $\gamma \rightarrow +\infty$,*

$$\frac{LO(G; \alpha, \beta, \gamma)}{e^{\gamma \text{imb}_{\max}(G)}} \longrightarrow \sum_{uv \in E_{\max}(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta,$$

and as $\gamma \rightarrow -\infty$, $LO(G; \alpha, \beta, \gamma)/e^{\gamma \text{imb}_{\min}(G)} \longrightarrow \sum_{uv \in E_{\min}(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta$.

Proof. At $\gamma = 0$ the exponential factor is 1 on every edge. For the asymptotics, write $M(G) := \text{imb}_{\max}(G)$ and $\varepsilon_{uv} := M(G) - \text{imb}(d_u, d_v) \geq 0$, so that $LO(G; \alpha, \beta, \gamma)/e^{\gamma M(G)} = \sum_{uv} (d_u d_v)^\alpha (d_u + d_v)^\beta e^{-\gamma \varepsilon_{uv}}$. For $uv \in E_{\max}(G)$, $\varepsilon_{uv} = 0$ and the summand is γ -independent; for $uv \notin E_{\max}(G)$, $\varepsilon_{uv} > 0$ and $e^{-\gamma \varepsilon_{uv}} \rightarrow 0$ as $\gamma \rightarrow +\infty$. Since $E(G)$ is finite the limit is termwise, giving the identity. The $\gamma \rightarrow -\infty$ case is dual. $\blacksquare$

5 Sensitivity analysis

We benchmark the Loyola family against the classical indices it reduces to, on three standard testbeds, following the chemometric benchmark of the diminished Sombor study [28]: the 18 octane isomers (five NIST physico-chemical properties), the 106 non-isomorphic trees of order 10, and the 75 decane isomers. The comparison set is restricted to the exact reductions of Section 1 — $M_1, M_2, HM, {}^mM_2, R, \chi, H/2, ISI, GA, AG$ — together with the two fixed pure- γ benchmark points $LO(0, 0, 1)$ and $LO(0, 0, 2)$ chosen for this study. No parameter in this comparison is tuned against the property data — the effect of data-driven tuning is examined separately, as a negative control, in Section 5.2; the edge count $m = LO(0, 0, 0)$ is omitted as it is constant on the octanes. All comparisons in this section are descriptive; given the number of properties, indices, and tuning candidates examined, no confirmatory inference is claimed.

5.1 Property correlations on the octane isomers

Table 1 reports the Pearson correlation r between each index and each of the five physicochemical properties of the 18 octane isomers (boiling point T_B , enthalpy of formation ΔH_f , enthalpy of vaporization ΔH_{vap} , entropy S , acentric factor ω ; property data from the NIST WebBook [23], compiled for internal consistency and comparison with diminished-Sombor-style benchmarks, accessed 2024–2025). Units are: T_B in $^{\circ}\text{C}$; ΔH_f and ΔH_{vap} in kcal mol^{-1} (the latter at 298 K); S in $\text{cal mol}^{-1}\text{K}^{-1}$; and ω dimensionless. Values for ΔH_{vap} and entropy S vary slightly across compilations; all comparisons here are made within this single, internally consistent dataset. Calculations use this author-compiled property table attributed to the NIST WebBook. All 90 molecule-level values were traced against authoritative sources (NIST WebBook first; KDB-derived compilations for the acentric factor, which NIST does not list), and classified under a strict taxonomy in which “verified” requires agreement at the value’s reported precision: 3 match exactly, 34 match at reported precision after unit conversion (gas phase confirmed for formation enthalpies and entropies), 32 agree within the audit tolerances or the source’s stated uncertainty without matching at reported precision, 13 lie within explained variation among authoritative determinations, 7 could not be verified against a reachable authoritative value (six gas-phase entropies absent from the NIST WebBook, and the 298 K vaporization enthalpy of 2,2,3,3-tetramethylbutane, which is solid at room temperature so that sublimation rather than vaporization data are tabulated), and one value is a marginal conflict: the acentric factors used here for 2,2,4-trimethylpentane (0.305 vs. compiled 0.303) and 2,2,3,3-tetramethylbutane (0.247 vs. compiled 0.251) differ from the KDB compilation by up to 0.004. Substituting the alternative compiled acentric factors changes several of the ω correlations in Table 1 at the third decimal (e.g. M_2 moves from -0.988 to -0.986) but does not change the property-wise winning descriptor or any substantive interpretation; the full sensitivity computation and the machine-readable provenance trace accompany the reproducibility package. The benchmark values themselves are preserved as supplied for comparability.

Table 1. Pearson correlation r between the Loyola reductions, the fixed pure- γ benchmark points and the five physicochemical properties of the 18 octane isomers. Bold flags the best $|r|$ per column.

Index	T_B	ΔH_f	ΔH_{vap}	S	ω
$M_1 = LO(0, 1, 0)$	-0.718	-0.762	-0.912	-0.973	-0.970
$M_2 = LO(1, 0, 0)$	-0.497	-0.542	-0.772	-0.940	-0.988
$HM = LO(0, 2, 0)$	-0.654	-0.710	-0.873	-0.975	-0.982
${}^mM_2 = LO(-1, 0, 0)$	+0.855	+0.891	+0.937	+0.863	+0.794
$R = LO(-1/2, 0, 0)$	+0.819	+0.850	+0.954	+0.933	+0.898
$\chi = LO(0, -1/2, 0)$	+0.800	+0.832	+0.951	+0.949	+0.924
$H/2 = LO(0, -1, 0)$	+0.821	+0.849	+0.956	+0.933	+0.900
$ISI = LO(1, -1, 0)$	-0.005	-0.000	-0.331	-0.601	-0.738
$GA = 2 LO(1/2, -1, 0)$	+0.822	+0.858	+0.957	+0.941	+0.906
$AG = \frac{1}{2} LO(-1/2, 1, 0)$	-0.816	-0.859	-0.950	-0.939	-0.900
$LO(0, 0, 1)$	-0.828	-0.832	-0.966	-0.931	-0.929
$LO(0, 0, 2)$	-0.829	-0.846	-0.967	-0.939	-0.925

The best correlation per property is split among the reductions and the pure- γ points: the modified second Zagreb index $LO(-1, 0, 0)$ is strongest on T_B and ΔH_f , the hyper-Zagreb index $LO(0, 2, 0)$ on S , the second Zagreb index $LO(1, 0, 0)$ on ω , and the fixed pure- γ benchmark point $LO(0, 0, 2)$ on ΔH_{vap} ($|r| = 0.967$). At the exact-reduction triples the Loyola correlations coincide with the corresponding classical indices to floating-point precision, as they must. For each property, a descriptive post-selection paired bootstrap interval for the observed top-two $|r|$ difference ($B = 10\,000$, resampling the 18 isomers jointly) included zero; these intervals were not multiplicity-adjusted across the five properties and do not establish equality or equivalence. Individual bootstrap half-widths range from about 0.02 to 0.10 across the headline cells. The value of $LO(0, 0, 2)$ on ΔH_{vap} is a fixed benchmark point, not a tuned result. Recomputing each bolded correlation with single isomers deleted shifts $|r|$ by at most 0.027, indicating that no single isomer drives the leading correlations.

Figure 1 shows the signed- r heatmap of the compared indices against the five properties.

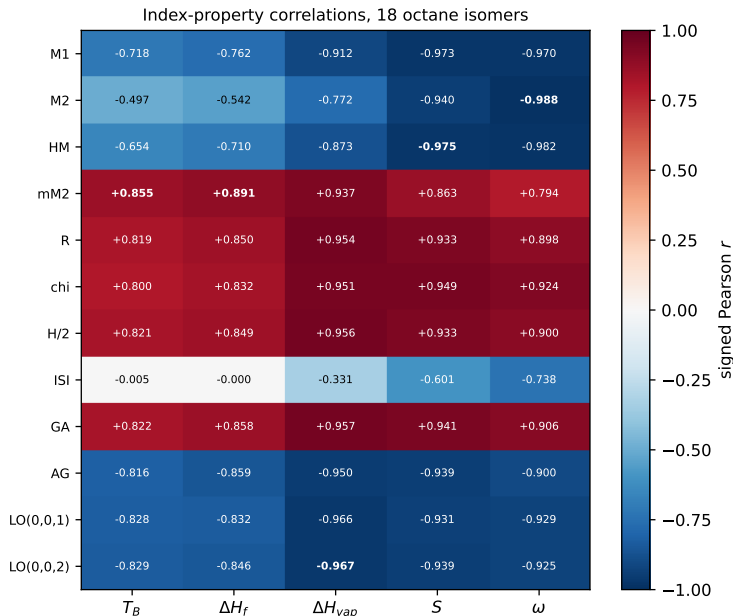


Figure 1. Signed Pearson r between the Loyola reductions, the fixed pure- γ benchmark points and the five octane physicochemical properties on the 18 isomers.

On the octanes the pure- γ points are highly collinear with the classical reductions (maximum $|r|$ with a reduction equal to 0.982 for $LO(0,0,1)$ and 0.993 for $LO(0,0,2)$).

Descriptor redundancy. The redundancy is in fact systemic, and it caps how much meaning any small $|r|$ difference in Table 1 can carry. Across the twelve compared descriptors on the 18 isomers, fourteen descriptor pairs have $|r| \geq 0.99$ (the extreme being R versus $H/2$ at $r = 0.9997$), and a principal component analysis of the standardized 18×12 descriptor matrix places 90.3% of the variance on the first component and 99.05% on the first two, with an effective rank (participation ratio of the eigenvalue spectrum) of 1.2. The twelve-index comparison set is thus essentially a two-dimensional descriptor space, and differences in the third decimal of $|r|$ between near-collinear descriptors should not be read as meaningful

rankings.

The information ceiling realised on this benchmark. Proposition 4 is not hypothetical on these data: the 18 isomers realise only 16 distinct edge-degree-pair count vectors, and the two collision pairs of Table 2 are indistinguishable by *every* vertex-degree-based index at *every* parameter choice, while their measured properties differ (by up to 1.2°C in T_B and 1.94 units in S within the first pair, and 2.1°C in T_B within the second). Every correlation in Table 1, and every tuned variant in the sections below, operates above this irreducible floor.

Table 2. The two degree-pair collision pairs among the 18 octane isomers. Each pair shares its complete edge-degree-pair count vector (shown as multiplicities of the unordered pairs), hence is indistinguishable by every vertex-degree-based index for all parameter values, yet differs in the measured properties.

Isomer	T_B	ΔH_f	ΔH_{vap}	S	ω
<i>shared count vector</i> 2(1, 2), 1(1, 3), 2(2, 2), 2(2, 3):					
3-methylheptane	118.9	-50.82	9.52	111.26	0.371
4-methylheptane	117.7	-50.69	9.48	109.32	0.372
<i>shared count vector</i> 2(1, 2), 2(1, 3), 2(2, 3), 1(3, 3):					
3,4-dimethylhexane	117.7	-50.91	9.32	106.59	0.340
2-methyl-3-ethylpentane	115.6	-50.48	9.21	106.06	0.330

5.2 Effect of parameter tuning near the best indices

The benchmark of Section 5.1 tunes no parameters: it compares only the exact reductions and the two fixed pure- γ benchmark points. It is nonetheless natural to ask whether small, data-driven adjustments of (α, β, γ) near the best-correlating indices could improve the fit. We probe this directly as a negative control. For each of the four anchors carrying the strongest column correlations in Table 1 — $M_2 = LO(1, 0, 0)$, $HM = LO(0, 2, 0)$, $^mM_2 = LO(-1, 0, 0)$, and the pure- γ point $LO(0, 0, 1)$ — we draw 50 random parameter triples within ± 0.3 (to two decimal places) of the anchor and correlate each against the five octane properties. Table 3 reports, for every anchor–property cell, the anchor correlation $|r|_0$, the best of the 50 perturbations in sample $|r|_{50}$, the in-sample gain Δ_{in} , the percentile-

bootstrap half-width ($B = 10000$), the leave-one-out correlation of the anchor LOO_0 , and a *nested* leave-one-out correlation LOO_{50} in which, within each of the four globally selected anchor pools, the best of the 50 candidates is selected using the 17 training isomers of each fold only and evaluated on the held-out one, with the resulting out-of-sample gain Δ_{out} . The anchors themselves were chosen from the complete-data correlations of Table 1, and the 18 pooled fold predictions share heavily overlapping training sets, so these values are descriptive rather than independent replications.

Table 3. Decimal parameter tuning near the four best-correlating anchors on the 18 octane isomers (50 random triples within ± 0.3 of each anchor). Columns: anchor correlation $|r|_0$; best-of-50 in sample $|r|_{50}$; in-sample gain Δ_{in} ; percentile-bootstrap half-width $\pm\text{boot}$ ($B = 10000$); leave-one-out correlation of the anchor LOO_0 ; foldwise nested leave-one-out correlation of the tuned family, computed within the globally selected anchor pool, LOO_{50} ; out-of-sample gain $\Delta_{\text{out}} = \text{LOO}_{50} - \text{LOO}_0$.

Property	$ r _0$	$ r _{50}$	Δ_{in}	$\pm\text{boot}$	LOO_0	LOO_{50}	Δ_{out}
<i>near</i> $M_2 = \text{LO}(1, 0, 0)$							
T_B	0.497	0.598	+0.100	0.353	0.331	0.458	+0.127
ΔH_f	0.542	0.635	+0.093	0.358	0.395	0.540	+0.145
ΔH_{vap}	0.772	0.839	+0.068	0.231	0.714	0.786	+0.073
S	0.940	0.960	+0.021	0.067	0.919	0.943	+0.024
ω	0.988	0.995	+0.007	0.018	0.985	0.985	0.000
<i>near</i> $HM = \text{LO}(0, 2, 0)$							
T_B	0.654	0.740	+0.085	0.257	0.545	0.647	+0.102
ΔH_f	0.710	0.786	+0.076	0.255	0.645	0.741	+0.097
ΔH_{vap}	0.873	0.920	+0.047	0.122	0.837	0.894	+0.057
S	0.975	0.976	+0.002	0.027	0.967	0.960	-0.007
ω	0.982	0.989	+0.007	0.025	0.979	0.982	+0.003
<i>near</i> ${}^mM_2 = \text{LO}(-1, 0, 0)$							
T_B	0.855	0.864	+0.009	0.096	0.781	0.759	-0.021
ΔH_f	0.891	0.897	+0.006	0.087	0.865	0.837	-0.028
ΔH_{vap}	0.937	0.966	+0.029	0.051	0.914	0.953	+0.039
S	0.863	0.918	+0.055	0.188	0.812	0.881	+0.069
ω	0.794	0.881	+0.088	0.246	0.730	0.853	+0.123
<i>near</i> $\text{LO}(0, 0, 1)$							
T_B	0.828	0.835	+0.007	0.161	0.771	0.764	-0.007
ΔH_f	0.832	0.842	+0.010	0.152	0.800	0.807	+0.006
ΔH_{vap}	0.966	0.969	+0.003	0.042	0.956	0.957	+0.002
S	0.931	0.948	+0.017	0.045	0.892	0.920	+0.028
ω	0.929	0.945	+0.016	0.074	0.907	0.926	+0.019

Three patterns emerge. First, every in-sample gain Δ_{in} is smaller than the corresponding single-descriptor bootstrap uncertainty scale; this comparison is descriptive and is not a formal paired or post-selection test of

the gain. Second, under the foldwise nested evaluation (within the globally selected anchor pools) these descriptive point-estimate gains do not vanish — tuning near M_2 raises the out-of-sample correlation for T_B and ΔH_f by +0.13 and +0.15, and tuning near mM_2 raises ω by +0.12 — but in every such case the anchor was a *weak* descriptor for that property, and the tuned correlation only climbs part way toward the level a strong reduction already attains (tuning near M_2 lifts T_B to 0.46, whereas mM_2 already reaches 0.78). Where the anchor is already the column winner — M_2 on ω , HM on S — the nested gain is zero or slightly negative. Third, tuning near the pure- γ points $LO(0,0,1)$ and $LO(0,0,2)$ does not exceed the best fixed point estimate in the tested comparison set on four of the five properties (Table 4). On ΔH_{vap} the tuned value exceeds the fixed benchmark $LO(0,0,2)$ by 0.002 in sample (0.969 vs 0.967) and by 0.003 under leave-one-out (0.959 vs 0.956); these small post-selection differences do not establish superiority. On T_B , ΔH_f , S , and ω the best fixed reduction is higher both in and out of sample.

Table 4. Decimal tuning near the pure- γ points $LO(0,0,1)$ and $LO(0,0,2)$ versus the best fixed index in the tested comparison set for each octane property. The best fixed index is ranked by in-sample $|r|$ (its leave-one-out value in parentheses); the decimal columns give the best of 50 random triples within ± 0.3 of either pure- γ point, in sample and under foldwise nested leave-one-out within the globally selected pools. Indices are ranked by in-sample $|r|$ rather than LOO, as the latter is inflated by degenerate fits for near-zero-correlation indices. Tuning does not exceed the best fixed point estimate on four properties; on ΔH_{vap} it exceeds it by 0.002 in sample and 0.003 under LOO, small post-selection differences that do not establish superiority.

Property	best fixed	$ r _{\text{fix}}$	LOO_{fix}	$ r _{\text{dec}}$	LOO_{dec}
T_B	mM_2	0.855	0.781	0.834	0.763
ΔH_f	mM_2	0.891	0.865	0.850	0.809
ΔH_{vap}	$LO(0,0,2)$	0.967	0.956	0.969	0.959
S	HM	0.975	0.967	0.949	0.921
ω	M_2	0.988	0.985	0.946	0.929

In short, local tuning raised several nested leave-one-out point estimates for weak anchors, and except for the small ΔH_{vap} margins noted

above did not exceed the best fixed benchmark in the tested comparison set; no formal superiority claim is made. All tabled nested values correspond to positive signed prediction–observation correlations with positive predictive Q^2 , and a constant-descriptor control confirms that absolute pooled correlation alone can be inflated for degenerate descriptors (cf. the caveat in the caption of Table 4).

5.3 Does the imbalance parameter add predictive information? A matched-budget nested comparison with the parent family

The analyses above tune around fixed anchors and are descriptive. The sharper question raised by the Gutman–Milovanović framing is direct: given identical search budgets, does freeing γ improve out-of-sample prediction beyond an optimized parent $M_{\alpha,\beta}$? We answer it with a fully nested leave-one-out design. For each property and each outer fold (one isomer held out), 200 candidate parameter vectors are drawn once from $\text{Uniform}(-2, 2)$ per coordinate (triples (α, β, γ) for the Loyola model, pairs (α, β) with $\gamma = 0$ for the parent; identical budget, fixed random seed); on the 17 training isomers only, every candidate is scored by *inner* leave-one-out root-mean-square error of the one-descriptor linear regression, the best candidate is selected, the regression is refit on all 17, and the held-out isomer is predicted. No quantity computed on the held-out isomer enters selection or fitting at any point.

Table 5. Matched-budget nested leave-one-out comparison of the optimized parent $M_{\alpha,\beta}$ (GM) and the optimized Loyola family (γ free) on the five octane properties. Q^2 is the outer predictive coefficient, RMSE the outer root-mean-square error; the last column counts outer folds (of 18) in which the Loyola model had the strictly smaller absolute error. All selection is repeated inside every outer fold.

Property	Q^2_{GM}	Q^2_{LO}	ΔQ^2	RMSE _{GM}	RMSE _{LO}	LO better
T_B	+0.613	+0.748	+0.135	3.700	2.983	10/18
ΔH_f	+0.855	+0.549	-0.306	0.463	0.816	4/18
ΔH_{vap}	+0.859	+0.935	+0.076	0.137	0.093	12/18
S	+0.915	+0.877	-0.038	1.310	1.573	4/18
ω	+0.979	+0.957	-0.022	0.005	0.007	5/18

The answer is mixed and, on balance, negative (Table 5): freeing γ improves nested predictive performance on two of the five properties (T_B , $\Delta Q^2 = +0.135$, and ΔH_{vap} , $\Delta Q^2 = +0.076$) and degrades it on three (ΔH_f , S , ω), with the ΔH_f degradation driven by selection instability: the parent model selected the same (α, β) in all 18 folds there, while the three-parameter search selected widely varying triples (fold-to-fold standard deviation above 1.2 in both α and β). The selected γ values are always interior to the search box (maximum $|\gamma| = 0.47$, never within 0.5 of the boundary), so the mixed outcome is not a range artefact. On the one property where γ helps most stably, ΔH_{vap} , the identical triple $(-0.27, +0.27, +0.24)$ was selected in every fold. With $n = 18$ molecules these Q^2 differences carry substantial selection noise and we attach no inferential claim to them.

Because the comparison depends on a random candidate stream, the full nested design was repeated for one hundred seeds (0–99) at both budgets (200 and 500 candidates per model; 1000 configurations in all), together with a paired control in which the Loyola candidates are compared against the *same* (α, β) coordinates with γ set to zero, isolating the effect of γ from the independent candidate streams. Freeing γ *usually* improved nested predictive performance for ΔH_{vap} (91/100 seeds at budget 200, 93/100 at budget 500; the paired control gives 93/100 at both budgets) and *usually* degraded it for ΔH_f (86/100 and 76/100 seeds), but neither direction is sign-invariant across random candidate streams; T_B is close

to an even split and S and ω usually degrade slightly. These are descriptive tendencies under the tested random-search procedure, not inferential evidence of predictive superiority or degradation. The honest summary is therefore: at matched budget, the imbalance parameter tends to help on one property (ΔH_{vap}) and tends to be unhelpful elsewhere, with no sign-invariant effect in either direction. Its value lies in the structural properties established in Section 3 and the discrimination behaviour below, not in systematic predictive gains.

5.4 Degeneracy across tree orders

Degeneracy measures an index’s ability to discriminate non-isomorphic graphs [21]; lower is better. On the 106 non-isomorphic trees of order 10 we use the distinct-value convention

$$\text{Degeneracy}(\%) = 100 \left(1 - \frac{k}{N} \right), \quad k = \#\{\text{distinct index values}\}, \quad N = 106,$$

i.e. the fraction of the N structures that fail to receive a unique value. The pure- γ points $LO(0, 0, 1)$ and $LO(0, 0, 2)$ attain degeneracy 25.5% (79 distinct values), among the lowest of all compared indices and tied with R , GA , and AG ; indeed the 106 trees realise only 79 distinct edge-degree-pair count vectors, so 25.5% is the minimum degeneracy attainable by *any* degree-pair-based index on this set, and the pure- γ points attain this floor — for vertex-degree-based weights φ_{ij} that are algebraic numbers, Rada [30] proved that the exponential index e^φ discriminates the equivalence classes of the corresponding $\widehat{m}_\varphi$ -structure relation; the pure- γ imbalance weights used here are rational, hence algebraic, so that theorem applies, and attainment of the 79-profile floor on this particular order-10 tree set is the separate computational observation reported here; the first Zagreb index $LO(0, 1, 0)$ (83.0%) and the second Zagreb index $LO(1, 0, 0)$ (67.0%) are the most degenerate (Figure 2).

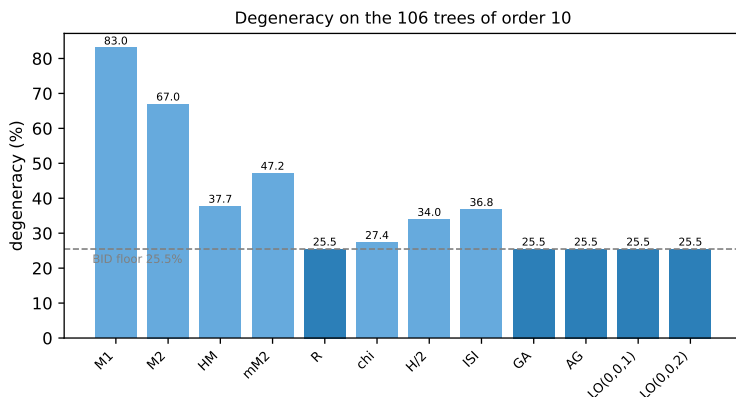


Figure 2. Degeneracy percentage of the compared indices on the 106 non-isomorphic trees of order 10.

To test whether this floor attainment is peculiar to order 10, the same computation was repeated for all non-isomorphic trees of orders 7 through 12 and, separately, for the molecular subsets with $\Delta \leq 4$ (Table 6). The pure- γ points attain the degree-pair-profile floor at *every* order in both sets. Honesty requires noting that the Randić index also attains the floor at every order in this range (while χ falls off the floor from order 10), so floor attainment distinguishes the pure- γ points from the most degenerate classical indices (M_1 , M_2) but not from R ; the observation is that the imbalance weights lose nothing, not that they are unique in this respect. Attainment on orders 7–12 is a computational observation, not a proof of injectivity on degree-pair profiles for all orders.

Table 6. Discrimination across tree orders $n = 7$ –12: number of trees N , number of distinct edge-degree-pair profiles (the floor attainable by any vertex-degree-based index), and distinct-value counts for representative indices. “mol.” denotes the molecular subset $\Delta \leq 4$.

n	set	N	floor	M_1	M_2	R	$LO(0, 0, 1)$	$LO(0, 0, 2)$
7	all	11	11	7	9	11	11	11
7	mol.	9	9	5	7	9	9	9
8	all	23	21	9	17	21	21	21
8	mol.	18	16	6	13	16	16	16
9	all	47	40	13	22	40	40	40
9	mol.	35	28	7	16	28	28	28
10	all	106	79	18	35	79	79	79
10	mol.	75	49	8	21	49	49	49
11	all	235	152	21	43	152	152	152
11	mol.	159	83	9	25	83	83	83
12	all	551	294	27	57	294	294	294
12	mol.	355	137	10	28	137	137	137

5.5 Structure sensitivity on decane isomers

The structure-sensitivity criterion for descriptor quality was introduced by Furtula, Gutman and Dehmer [9], with a fingerprint-based variant due to Rakić and Furtula [32]. We implement the published protocol on the 75 decane isomer trees Ψ : for each tree G , the similar-structure set $S(G) \subset \Psi$ consists of the non-isomorphic trees obtainable from G by relocating a single edge within the molecular class (graph edit distance two, $\Delta \leq 4$ preserved); then

$$SS(TI, G) = \frac{1}{|S(G)|} \sum_{\Omega \in S(G)} \frac{|TI(\Omega) - TI(G)|}{TI(G)},$$

$$Abr(TI, G) = \max_{\Omega \in S(G)} \frac{|TI(\Omega) - TI(G)|}{TI(G)},$$

and the class-level $SS(TI)$ and $Abr(TI)$ are the means over all 75 trees. On this class every $S(G)$ is nonempty (sizes 7 to 43), the neighbour relation is symmetric, and $Abr \geq SS$ holds throughout, as it must. As an external control, the implementation reproduces the published decane-class values of Barman and Das [4] for the seven indices common to both studies (M_1 ,

mM_2 , R , χ , H , GA , AG ; fourteen tabulated SS/Abr values): ten agree exactly at the four displayed decimals and the remaining four differ by one unit in the fourth decimal. Under the published desideratum (SS large, Abr small [9]), the pure- γ point $LO(0, 0, 2)$ attains the largest structure sensitivity in the comparison set ($SS = 0.1257$, marginally above the hyper-Zagreb index at 0.1255) with a markedly smaller abruptness than hyper-Zagreb (0.274 versus 0.313), and the two pure- γ points have the two largest sensitivity-to-abruptness ratios (0.4579 and 0.4565). Given the redundancy analysis of Section 5.1, the SS margin over hyper-Zagreb should be read as a tie in sensitivity at smaller abruptness, not as a ranking.

Table 7. Structure sensitivity SS , abruptness Abr , and their ratio on the 75 decane isomers under the published similar-structure protocol of [9] (single-edge relocation within the molecular class).

Index	SS	Abr	SS/Abr
M_1	0.0566	0.1336	0.4238
M_2	0.0922	0.2443	0.3776
HM	0.1255	0.3126	0.4017
mM_2	0.0516	0.1177	0.4384
R	0.0264	0.0588	0.4481
χ	0.0264	0.0591	0.4463
$H/2$	0.0516	0.1143	0.4514
ISI	0.0226	0.0597	0.3795
GA	0.0241	0.0536	0.4507
AG	0.0256	0.0579	0.4415
$LO(0, 0, 1)$	0.0624	0.1368	0.4565
$LO(0, 0, 2)$	0.1257	0.2745	0.4579

Earlier drafts of this study reported the coefficient of variation σ/μ and relative range $(\max - \min)/\mu$ over the isomer class under the names SS and Abr ; those are global dispersion summaries, not the published pairwise quantities, and they are retained only as internal artefacts of the reproducibility package under the names *normalized dispersion* and *normalized range*.

6 Conclusion and further directions

The Loyola family is the normalized degree-imbalance exponential extension of the Gutman–Milovanović family $M_{\alpha,\beta}$ [13, 26], to which it reduces identically at $\gamma = 0$. What the third parameter contributes is now stated precisely: the imbalance direction is affinely independent of the two parent directions on graphs with $\Delta \geq 3$ (Proposition 2); the family has a globally convex logarithm whose γ -derivatives are the mean and variance of the edge imbalance under an induced edge measure (Theorem 3); and the two-sided bounds of Section 4 extend inequalities known for the parent to the $\gamma \neq 0$ regime, several of them sharp.

Chemometrically the conclusions are deliberately honest. Every vertex-degree-based index on the octane benchmark operates above an irreducible floor set by two degree-pair collision pairs (Proposition 4); the twelve-index comparison set is effectively two-dimensional, so third-decimal correlation differences carry no ranking information; and at matched search budget, across one hundred random-search seeds and two budgets, the free imbalance parameter usually improves nested out-of-sample prediction on one of five properties and usually does not elsewhere, with neither tendency sign-invariant across candidate streams (Table 5) — it is not a systematic predictive upgrade over the optimized parent. Its measurable contribution is structural: the pure- γ points attain the degree-pair discrimination floor at every tree order 7–12 tested, and under the published structure-sensitivity protocol they combine hyper-Zagreb-level sensitivity with smaller abruptness and the largest sensitivity-to-abruptness ratios in the comparison set. The Loyola family is hence best viewed as a provably independent, analytically transparent third direction on the Gutman–Milovanović parameter space, confined — like every vertex-degree-based family — to the finite-dimensional degree-pair descriptor space on molecular graphs.

Building on the Loyola index, the following open problems indicate natural directions for further work:

1. Sharp two-sided bounds for $LO(G; \alpha, \beta, \gamma)$ via degree-based indices lying outside the family, such as the atom-bond-connectivity index ABC and the Sombor index.

2. Extremal-graph characterisations for several graph classes, including trees, unicyclic and bicyclic graphs.
3. Chemometric extension to additional chemical compound classes such as benzene derivatives, alkane homologues beyond octane and decane and to compound-class-specific QSPR endpoints.

Acknowledgment: The authors thank colleagues for helpful discussions of the computational extremal-graph search.

A Octane isomer property data

For reproducibility, Table 8 lists the full 18-isomer dataset used in Section 5. Units: T_B in $^{\circ}\text{C}$; ΔH_f and ΔH_{vap} in kcal mol^{-1} (ΔH_{vap} at 298 K); S in $\text{cal mol}^{-1}\text{K}^{-1}$; ω dimensionless.

Table 8. Physicochemical properties of the 18 octane isomers.

Isomer	T_B	ΔH_f	ΔH_{vap}	S	ω
<i>n</i> -octane	125.6	-49.82	9.92	111.55	0.398
2-methylheptane	117.6	-51.5	9.48	109.84	0.378
3-methylheptane	118.9	-50.82	9.52	111.26	0.371
4-methylheptane	117.7	-50.69	9.48	109.32	0.372
2,2-dimethylhexane	106.8	-53.71	8.92	103.13	0.339
2,3-dimethylhexane	115.6	-51.13	9.27	108.02	0.348
2,4-dimethylhexane	109.4	-52.44	9.03	106.98	0.344
2,5-dimethylhexane	109.1	-53.21	9.05	105.72	0.357
3,3-dimethylhexane	111.9	-52.61	9.04	104.74	0.322
3,4-dimethylhexane	117.7	-50.91	9.32	106.59	0.34
2-methyl-3-ethylpentane	115.6	-50.48	9.21	106.06	0.33
3-ethyl-3-methylpentane	118.3	-51.38	9.21	101.48	0.302
3-ethylhexane	118.5	-50.4	9.48	109.43	0.362
2,2,3-trimethylpentane	109.8	-52.61	8.88	101.31	0.3
2,2,4-trimethylpentane	99.2	-53.57	8.4	101.81	0.305
2,3,3-trimethylpentane	114.8	-51.73	9.02	101.31	0.291
2,3,4-trimethylpentane	113.5	-51.97	9.01	102.39	0.317
2,2,3,3-tetramethylbutane	106.5	-53.99	8.41	93.06	0.247

References

- [1] S. Ahmad, R. Farooq, K. C. Das, The general Sombor index of extremal trees with a given maximum degree, *MATCH Commun. Math. Comput. Chem.* **94** (2025) 825–853.
- [2] M. O. Albertson, The irregularity of a graph, *Ars Combin.* **46** (1997) 219–225.
- [3] M. Arockiaraj, D. Paul, J. Clement, S. Tigga, K. Jacob, K. Balasubramanian, Novel molecular hybrid geometric-harmonic-Zagreb degree based descriptors and their efficacy in QSPR studies of polycyclic aromatic hydrocarbons, *SAR QSAR Environ. Res.* **34** (2023) 569–589.
- [4] J. Barman, S. Das, Geometric approach to degree-based topological index: Hyperbolic Sombor index, *MATCH Commun. Math. Comput. Chem.* **95** (2026) 63–94.
- [5] B. Bollobás, P. Erdős, Graphs of extremal weights, *Ars Combin.* **50** (1998) 225–233.
- [6] K. C. Das, A. S. Çevik, I. N. Cangul, Y. Shang, On Sombor index, *Symmetry* **13** (2021) #140.
- [7] K. C. Das, I. Gutman, On Sombor index of trees, *Appl. Math. Comput.* **412** (2022) #126575.
- [8] B. Furtula, I. Gutman, A forgotten topological index, *J. Math. Chem.* **53** (2015) 1184–1190.
- [9] B. Furtula, I. Gutman, M. Dehmer, On structure-sensitivity of degree-based topological indices, *Appl. Math. Comput.* **219** (2013) 8973–8978.
- [10] W. Gao, Y. Gao, The extremal trees for exponential vertex-degree-based topological indices, *Appl. Math. Comput.* **472** (2024) 128634.
- [11] A. Granados, A. Portilla, Y. Quintana, E. Tourís, Bounds for the Gutman–Milovanović index and some applications, *J. Math. Chem.* **63** (2025) 406–418.
- [12] R. Guji, M. Wali, The Sombor index (coindex) and Lanzhou index (coindex) of some graphs, *Axioms* **14** (2025) 164.
- [13] I. Gutman, E. Milovanović, I. Milovanović, Beyond the Zagreb indices, *AKCE Int. J. Graphs Comb.* **17** (2020) 74–85.

-
- [14] I. Gutman, M. Togan, A. Yurttaş, A. S. Çevik, I. N. Cangul, Inverse problem for sigma index, *MATCH Commun. Math. Comput. Chem.* **79** (2018) 491–508.
- [15] I. Gutman, Geometric approach to degree-based topological indices: Sombor indices, *MATCH Commun. Math. Comput. Chem.* **86** (2021) 11–16.
- [16] I. Gutman, Some basic properties of Sombor indices, *Open J. Discr. Appl. Math.* **4** (2021) 1–3.
- [17] I. Gutman, B. Furtula, M. S. Oz, Geometric approach to vertex-degree-based topological indices: Elliptic Sombor index, theory and application, *Int. J. Quantum Chem.* **124** (2024) #e27346.
- [18] I. Gutman, Survey of Sombor-type degree-based topological indices, *MATCH Commun. Math. Comput. Chem.* **96** (2026) 819–837.
- [19] Z. Hu, L. Li, X. Li, D. Peng, Extremal graphs for topological index defined by a degree-based edge-weight function, *MATCH Commun. Math. Comput. Chem.* **88** (2022) 505–520.
- [20] A. R. A. Kalaam, A. B. Greeni, M. Arockiaraj, Modified reverse degree descriptors for combined topological and entropy characterizations of 2D metal organic frameworks: applications in graph energy prediction, *Front. Chem.* **12** (2024) #1470231.
- [21] E. V. Konstantinova, The discrimination ability of some topological and information distance indices for graphs of unbranched hexagonal systems, *J. Chem. Inf. Comput. Sci.* **36** (1996) 54–57.
- [22] S. Kosari, N. Dehghardi, A. Khan, Lower bound on the KG-Sombor index, *Commun. Comb. Optim.* **8** (2023) 751–757.
- [23] P. J. Linstrom, W. G. Mallard (Eds.), *NIST Chemistry WebBook, NIST Standard Reference Database Number 69*, National Institute of Standards and Technology, Gaithersburg MD, doi:10.18434/T4D303 (retrieved January 2025).
- [24] J. A. Méndez-Bermúdez, R. Aguilar-Sánchez, E. D. Molina, J. M. Rodríguez, Mean Sombor index, *Discr. Math. Lett.* **9** (2022) 18–25.
- [25] I. Milovanović, E. Milovanović, M. Matejić, On some mathematical properties of Sombor indices, *Bull. Int. Math. Virtual Inst.* **11** (2021) 341–353.

-
- [26] E. D. Molina, J. M. Rodríguez-García, J. M. Sigarreta, S. J. Torralbas Fitz, On the Gutman–Milovanović index and chemical applications, *AIMS Math.* **10** (2025) 1998–2020.
- [27] F. Movahedi, Diminished Sombor index and its relationship with topological indices, *Filomat* **39** (2025) 10519–10532.
- [28] F. Movahedi, I. Gutman, I. Redžepović, B. Furtula, Diminished Sombor index, *MATCH Commun. Math. Comput. Chem.* **95** (2026) 141–162.
- [29] A. Portilla, J. M. Rodríguez, I. Salas Lorenzo, J. M. Sigarreta, The Gutman–Milovanović index in mathematical chemistry, *J. Math. Chem.* **63** (2025) 2068–2087.
- [30] J. Rada, Exponential vertex-degree-based topological indices and discrimination, *MATCH Commun. Math. Comput. Chem.* **82** (2019) 29–41.
- [31] J. Rada, A finite-dimensional linear framework for vertex-degree-based topological indices, *MATCH Commun. Math. Comput. Chem.* **96** (2026) 841–863.
- [32] M. Rakić, B. Furtula, A novel method for measuring the structure sensitivity of molecular descriptors, *J. Chemometrics* **33** (2019) e3138.
- [33] Y. Shang, Sombor index and degree-related properties of simplicial networks, *Appl. Math. Comput.* **419** (2022) 126881.
- [34] J. M. Sigarreta, Extremal problems on exponential vertex-degree-based topological indices, *Math. Biosci. Eng.* **19** (2022) 6985–6995.
- [35] D. Vukičević, B. Furtula, Topological index based on the ratios of geometrical and arithmetical means of end-vertex degrees of edges, *J. Math. Chem.* **46** (2009) 1369–1376.
- [36] Z. Wang, Y. Mao, Y. Li, B. Furtula, On relations between Sombor and other degree-based indices, *J. Appl. Math. Comput.* **68** (2022) 1–17.
- [37] B. Zhou, N. Trinajstić, On a novel connectivity index, *J. Math. Chem.* **46** (2009) 1252–1270.
- [38] B. Zhou, N. Trinajstić, On general sum-connectivity index, *J. Math. Chem.* **47** (2010) 210–218.
- [39] X. Zuo, B. A. Rather, M. Imran, A. Ali, On some topological indices defined via the modified Sombor matrix, *Molecules* **27** (2022) 6772.